

KHAIRUNNISA KHAN BINTI TALIB

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[Portfolio: portfolio-nisa-eight.vercel.app](https://portfolio-nisa-eight.vercel.app) | [GitHub: github.com/nisakhantalib](https://github.com/nisakhantalib)

SUMMARY

Software engineer with production experience in React, Next.js, and TypeScript, and a machine-learning foundation from the University of Manchester (BSc Computer Science). Built a government digital-platform prototype used in a national tender submission, developed a full-stack AI learning platform integrating LLM APIs, and implemented CI-integrated Playwright test suites. AWS Certified Solutions Architect – Associate.

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Python, Java, SQL
- **Frontend:** React, Next.js (App Router), Tailwind CSS, Material UI (Material 3), HTML5, CSS3, Recharts, Framer Motion
- **Testing & Quality:** Playwright (UI, API, accessibility, regression), Test-Driven Development (TDD), GitHub Actions CI/CD
- **AI / Machine Learning:** PyTorch, TensorFlow, Keras, scikit-learn, BERT, Bi-LSTM, LLM API integration (Groq / Llama), Model Context Protocol (MCP), Hugging Face Transformers
- **Data & Databases:** MySQL, MongoDB, Oracle SQL, Power BI, data cleaning & visualization
- **Cloud & Tools:** AWS (certified), Vercel, Git/GitHub, n8n workflow automation, Flask, Spring Boot

WORK EXPERIENCE

Frontend Developer | Artech Niaga Sdn Bhd

Oct 2025 – Present

- Built a functional prototype of the MAQIS Digital Platform, a national quarantine and inspection management system, using Next.js, React, TypeScript, and Tailwind CSS; deployed on Vercel for stakeholder evaluation
- Developed the admin portal with authentication flows, bilingual (Malay/English) UI, and interactive Recharts dashboards for operational reporting
- Authored technical documentation and compliance materials that supported the company's government tender submission

Solution Engineering Intern | Infobip Solution Malaysia

Mar 2025 – Jun 2025

- Supported client engagement sessions, including live product demos, proof-of-concept presentations, and technical Q&A for enterprise messaging solutions
- Customized chatbot implementations to customer requirements, translating business needs into working conversational flows
- Prepared User Acceptance Testing (UAT) documentation to support client-side solution validation and sign-off

PROJECTS

Bangkit — AI-Powered Learning Platform — Next.js, React, Groq LLM API, React Flow, Tailwind CSS [Live Demo](#) | [GitHub](#)

- Engineered a full-stack exam-preparation platform for Malaysian SPM students, combining curriculum-aligned lessons, past-paper practice, an AI tutor, and LLM-generated quizzes with automated marking
- Designed a resilient LLM integration layer with multi-model fallback and provider cooldown across Groq-hosted models (Llama 3.3 70B), exposed through Next.js API routes for chat, quiz generation, and marking
- Visualized the syllabus as an interactive knowledge graph with React Flow, and implemented difficulty levels, notes, and progress tracking to personalize revision

SetupHub — E-commerce Test Automation Suite — Playwright, Node.js, Express, GitHub Actions [GitHub](#)

- Built an end-to-end Playwright automation suite for a Node/Express e-commerce application covering UI flows, API validation, regression scenarios, accessibility checks, and mocked network-failure conditions
- Integrated the suite into GitHub Actions CI with Playwright HTML reporting for repeatable, reviewable test runs on every push

Evidence Detection with NLP — Python, PyTorch, BERT, Bi-LSTM [GitHub](#)

- Trained and evaluated Bi-LSTM and BERT models for evidence detection, achieving 90% accuracy with both architectures and comparing trade-offs between recurrent and transformer approaches

Paws Preferences — Interactive Web App — React, Tailwind CSS, REST API [Live Demo](#) | [GitHub](#)

- Developed a responsive swipe-based image browsing application with dynamic API-driven content loading, deployed on GitHub Pages

EDUCATION

BSc (Hons) Computer Science — 2:1 (67.5%) University of Manchester, UK	2021 – 2024
Relevant coursework: Natural Language Processing, Machine Learning, Database Design (Oracle, SQL)	
Cambridge A Levels — Mathematics (A*), Physics (A*), Biology (A) Kolej Teknologi Timur	2019 – 2021
SPM: 9A (SMK Oya, Sarawak)	

CERTIFICATIONS

- **AWS Certified Solutions Architect – Associate** — Amazon Web Services (Jan 2025 – Jan 2028)
- **AWS Certified Cloud Practitioner** — Amazon Web Services (Jan 2025 – Jan 2028)
- **Google Data Analytics Professional Certificate** — Google
- **CompTIA Cloud Essentials+** — CompTIA
- **Test-Driven Development for JavaScript (Units 1 & 2)** — Pearson / Coursera (Apr 2026)
- **Advanced UI with Material 3: Customization & Performance** — Coursera (Mar 2026)
- **Workflow Automation with n8n: Logic, Data & Error Handling** — LearnKartS / Coursera (Jul 2026)
- **Introduction to Model Context Protocol** — Anthropic (2026)

ADDITIONAL

- **Languages:** English (IELTS Band 7.5), Malay (native), Arabic (basic)
- **Public speaking:** 2nd runner-up, Forum Remaja national final (2018); national-level parliamentary-style debate participant